One end of the semi elliptical spring is attached to the frame through shackle and the other end of the semi elliptical spring is attached to the quarter elliptical spring by means of the shackle. The other end is bolted to the frames and being rigidly held by I – bolt. This type of arrangement was more popular in olden days and not used now-a-days.

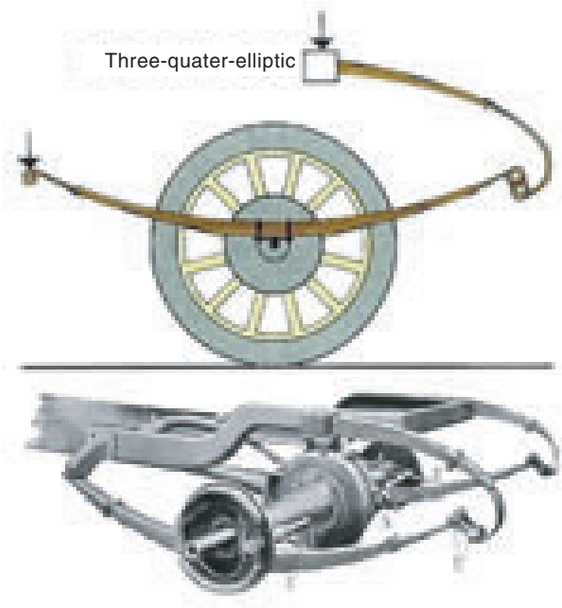


Figure 7.1.1.1 (c) Three quarter elliptic leaf spring

(d) Full Elliptic Leaf Spring

The simple construction of this type is shown in figure 7.1.1.1 (d). In this type, two semi elliptical springs are connected in opposite manner to form an elliptical shape. The full elliptical spring is attached rigidly to both the axle and the frame. Spring shackles are not necessary, since both the top and bottom members will elongate by the same amount when compressed. This arrangement was used in olden cars and not used now-a-days.

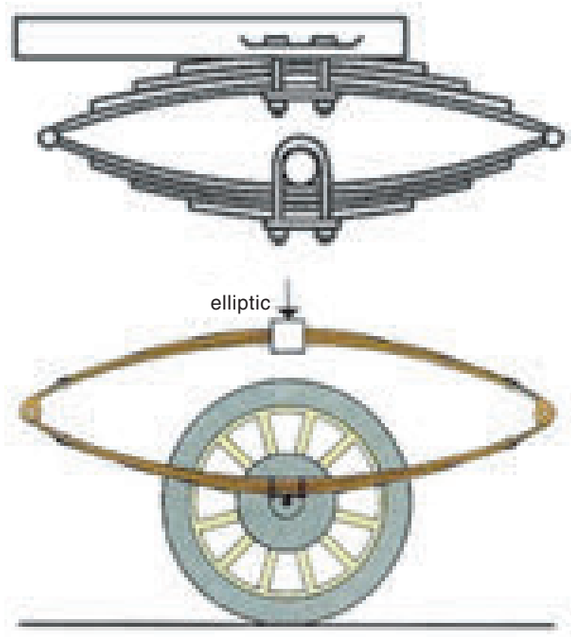


Figure 7.1.1.1 (d) Full elliptic leaf spring

(e) Transversely mounted semi-elliptical inverted leaf spring

The figure 7.1.1.1 (e) shows the arrangement. In this arrangement, a semi-elliptical leaf spring is mounted transversely along the width of the vehicle. The springs are placed, so inverted that the longest leaf is located at the bottom. The mid portion is fixed to the frame by means of U-bolt. The specialty of this arrangement is the use of two shackles. The rolling tendency of this type leads to its unsuitability for vehicles.

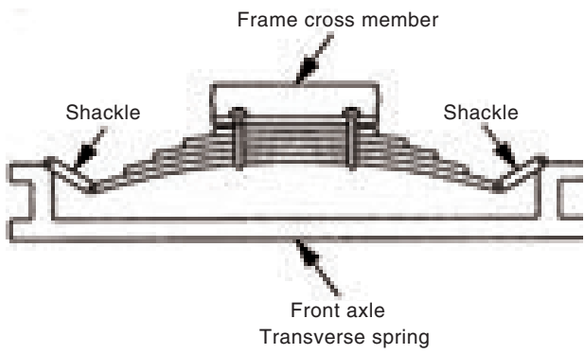


Figure 7.1.1.1 (e) Transversely mounted semi-elliptical inverted leaf spring

BAE_ENG_Chapter_07.indd 127 12/10/2019 11:54:02 AM